

V4 parameter estimation

One model · SPY, AAPL, MSFT, AMZN · four windows · 1.5-year monthly calibration

What this report is

- The rolling parameter paths used in this study, summarized by regime. One table per model.
- No LSM, no option prices, and no simulated paths. Only calibrate_ticker with the 18-month lookback.
- The companion notebook is the computational source of truth. This PDF is written from the same payload.

How estimates are formed

- At monthly update t , parameters use only returns and listed quotes with `trading_date ≤ t`. LSM prices a Monday contract with the latest calibration row dated on or before that Monday. Option quotes are as-of the grid timestamp (same session, else the prior session).
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- Cells are Mean, SD, Min, and Max of those monthly estimates inside the regime.

Tables

- P1 GARCH.
- R1 Crisis, R2 Normal, R3 Late-cycle, R4 COVID. Columns are SPY, AAPL, MSFT, AMZN.

Table P1 · GARCH · monthly parameter estimates

1.5-year lookback · rolling=monthly · Mean, SD, Min, Max within each regime

gime / para	SPY				AAPL				MSFT				AMZN			
gime / para	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis																
λ	-0.0247	0.0178	-0.0470	0.0104	0.0718	0.0302	0.0442	0.1274	-0.0162	0.0123	-0.0340	0.0058	0.0415	0.0234	0.0056	0.0800
ω	5.610e-06	1.728e-06	2.700e-06	8.327e-06	4.785e-05	1.437e-05	1.535e-05	6.194e-05	2.274e-05	8.998e-06	1.196e-05	4.171e-05	4.422e-04	4.626e-04	1.881e-05	9.889e-04
α	0.1113	0.0216	0.0553	0.1328	0.1226	0.0166	0.0991	0.1532	0.1617	0.0431	0.1023	0.2160	0.1583	0.1450	0.0461	0.4570
β	0.8787	0.0192	0.8539	0.9276	0.8334	0.0260	0.7917	0.8766	0.8202	0.0356	0.7726	0.8845	0.4938	0.4761	7.032e-54	0.9354
σ_0	0.0254	0.0127	0.0125	0.0501	0.0344	0.0191	0.0177	0.0867	0.0325	0.0126	0.0151	0.0549	0.0351	0.0083	0.0262	0.0583
R2 Normal																
λ	0.1362	0.0122	0.1186	0.1614	0.0647	0.0534	-0.0073	0.1415	0.1320	0.0461	0.0695	0.1863	0.0372	0.0308	0.0108	0.1122
ω	1.090e-05	3.206e-06	5.846e-06	1.521e-05	2.536e-06	4.270e-06	8.356e-43	1.192e-05	1.104e-04	3.669e-05	7.090e-05	1.662e-04	2.147e-04	8.497e-05	4.034e-05	2.967e-04
α	0.1600	0.0339	0.1222	0.2220	0.0083	0.0139	1.645e-129	0.0381	0.3529	0.1915	0.1078	0.6187	0.3203	0.1088	0.1213	0.4827
β	0.6275	0.0726	0.5147	0.7307	0.9842	0.0245	0.9365	0.9997	0.2580	0.0593	0.1836	0.3705	0.1081	0.2642	7.014e-15	0.7372
σ_0	0.0064	9.323e-04	0.0049	0.0083	0.0147	0.0025	0.0126	0.0216	0.0136	0.0016	0.0107	0.0167	0.0173	0.0038	0.0142	0.0291
R3 Late-cycle																
λ	0.1459	0.0280	0.1055	0.1865	0.1067	0.0194	0.0746	0.1370	0.1382	0.0150	0.1141	0.1720	0.1559	0.0249	0.1017	0.1899
ω	3.612e-06	1.256e-06	1.959e-06	5.785e-06	2.245e-05	9.679e-06	1.083e-05	4.363e-05	2.908e-05	9.041e-06	1.511e-05	4.244e-05	3.489e-05	1.797e-05	1.360e-05	6.055e-05
α	0.2097	0.0272	0.1665	0.2516	0.1526	0.0239	0.1146	0.1902	0.2085	0.0294	0.1355	0.2601	0.1956	0.0328	0.1293	0.2527
β	0.7631	0.0213	0.7216	0.7965	0.7692	0.0634	0.6206	0.8568	0.6706	0.0717	0.5159	0.7981	0.7202	0.0492	0.6160	0.8079
σ_0	0.0095	0.0052	0.0044	0.0230	0.0169	0.0060	0.0108	0.0288	0.0154	0.0051	0.0096	0.0269	0.0203	0.0097	0.0125	0.0413
R4 COVID																
λ	0.1571	0.0275	0.1093	0.1929	0.1688	0.0400	0.1151	0.2350	0.1396	0.0186	0.1141	0.1685	0.0929	0.0231	0.0502	0.1309
ω	4.488e-06	5.784e-07	3.298e-06	5.817e-06	1.705e-05	2.450e-06	1.309e-05	2.072e-05	1.271e-05	3.167e-06	7.421e-06	2.029e-05	1.614e-05	5.812e-06	1.074e-05	2.975e-05
α	0.2515	0.0569	0.1677	0.3259	0.1616	0.0301	0.1224	0.2273	0.1858	0.0367	0.1355	0.2400	0.1510	0.0324	0.1034	0.2019
β	0.7293	0.0386	0.6731	0.7890	0.8052	0.0268	0.7506	0.8388	0.7736	0.0305	0.7227	0.8060	0.8081	0.0443	0.7164	0.8659
σ_0	0.0142	0.0127	0.0053	0.0487	0.0210	0.0116	0.0121	0.0536	0.0201	0.0144	0.0092	0.0597	0.0177	0.0062	0.0119	0.0333

Notes: This table reports Mean, SD, Min, and Max of the monthly GARCH estimates for SPY, AAPL, MSFT, AMZN. Parameters are λ , ω , α , β , σ_0 . Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.